TRAIL RANGER ACADEMY · GRADE 7

Slope & Rate of Change — Homework

Name: _____

Date: _____

📓 Piper's Trail Journal: the Secret Message

After the Lantern Festival, Piper found a coded note pinned to the Summit Beacon. It says the **next adventure** starts where the code points! Solve the 14 locks below. Each answer matches **one letter** in the Decoder Table. Write the letter in the box and read the message at the end.

- Write every answer as a **whole number** or a **simplified fraction** (or the word *undefined*).
- If your answer is **not in the table**, check it again: it is a trap!
- Part B needs your full work and sentences.

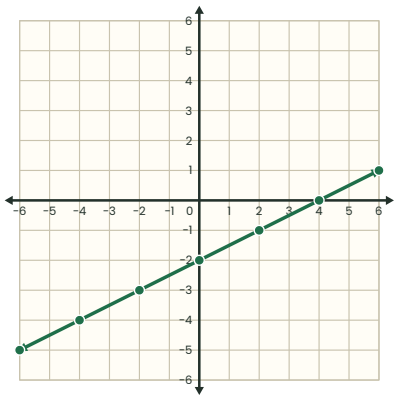
🔐 Decoder Table

3 → **X**	3/2 → **Z**	-3 → **O**	undefined → **W**	3/4 → **V**	4/3 → **G**
-1/4 → **E**	-4 → **K**	4 → **T**	5/3 → **R**	3/5 → **D**	2 → **S**
2/3 → **L**	5 → **I**	1/2 → **F**	0 → **H**	-2/3 → **Q**	-1/2 → **A**

Part A · The 14 Locks

1 Find the slope of the line.

Answer: _____ Letter: ☐



2 Find the slope of the line through **(-2, 7)** and **(1, -2)**.

Answer: _____ Letter: ☐

3 Find the slope shown in the table.

Answer: _____ Letter: ☐

x	0	3	6	9
y	1	3	5	7

4 A diver swims **up 8 meters** in **12 seconds**. What is the rate of change in meters per second?

Answer: _____ **m per s** Letter: ☐

5 A pile of snow melts from **24 cm** tall to **9 cm** tall in **5 hours**. What is the rate of change in cm per hour?

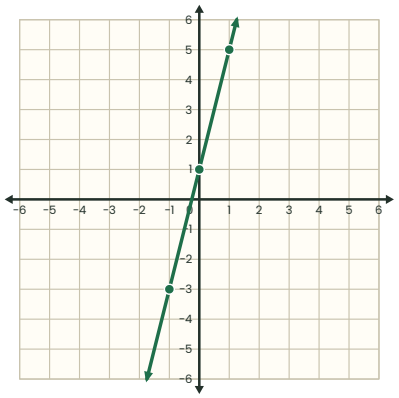
Answer: _____ **cm per h** Letter: ☐

6 Find the slope of the line through **(4, 7)** and **(4, -1)**. *(Write "undefined" if it has no slope.)*

Answer: _____ Letter: ☐

7 Find the slope of the line.

Answer: _____ Letter: ☐

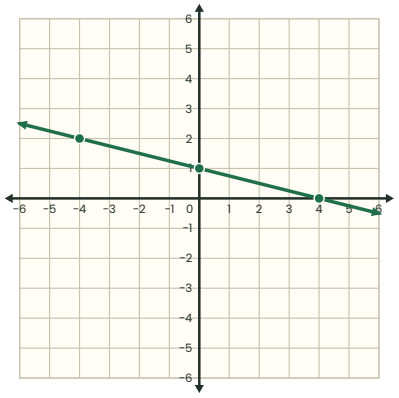


8 What is the slope of the line **y = -3**?

Answer: _____ Letter: ☐

9 Find the slope of the line.

Answer: _____ Letter: ☐



10 Find the slope of the line through **(-3, -4)** and **(3, 6)**.

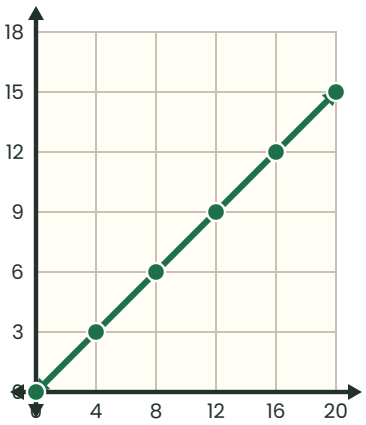
Answer: _____ Letter: ☐

11 A rocket climbs **150 meters** in **30 seconds**. What is the rate of change in meters per second?

Answer: _____ **m per s** Letter: ☐

12 Find the slope of the line. **Careful:** x counts by 4s and y counts by 3s.

Answer: _____ Letter: ☐



13 Find the slope shown in the table.

Answer: _____ Letter: ☐

x	0	4	8	12
y	9	8	7	6

14 A tank fills with **20 liters** of water in **12 minutes**. What is the rate of change in liters per minute?

Answer: _____ **L per min** Letter: ☐

📬 **Write the letters from locks 1–14 here, in order:**

☐☐☐☐☐☐ ☐☐☐ ☐☐☐☐☐

1 2 3 4 5 6 7 8 9 10 11 12 13 14

Part B · Show Your Thinking

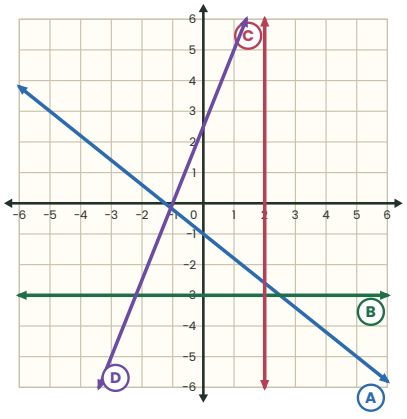
15 Name the **type of slope** for each line (positive, negative, zero, or undefined).

_____

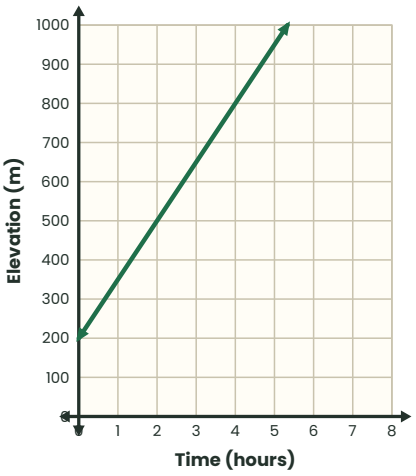
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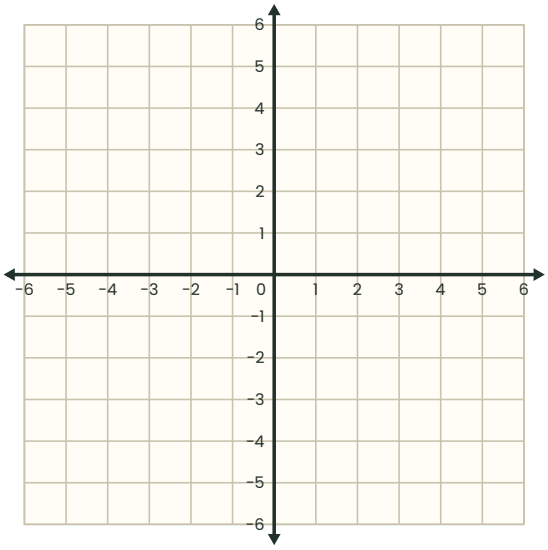


16 The graph shows a hiker's elevation over time. (a) Find the rate of change, with units. (b) Explain in a sentence what it means.



17 Sam finds the slope through **(2, 5)** and **(6, 1)** like this: (5 - 1) ÷ (6 - 2) = 4/4 = 1. Find Sam's mistake and give the correct slope.

18 Graph the line through **(1, 4)** with slope **−2/3**. Plot at least three points and draw the line. Show how you moved from the first point.



19 Car A's trip is in the table. Car B travels **240 miles in 4 hours**. Which car is faster, and by how many miles per hour?

Hours	1	2	3
Miles	55	110	165

20 In your own words, explain the difference between a **zero** slope and an **undefined** slope. Give a real-world example of each.

Check your work: the secret message should be three real words. 🧭